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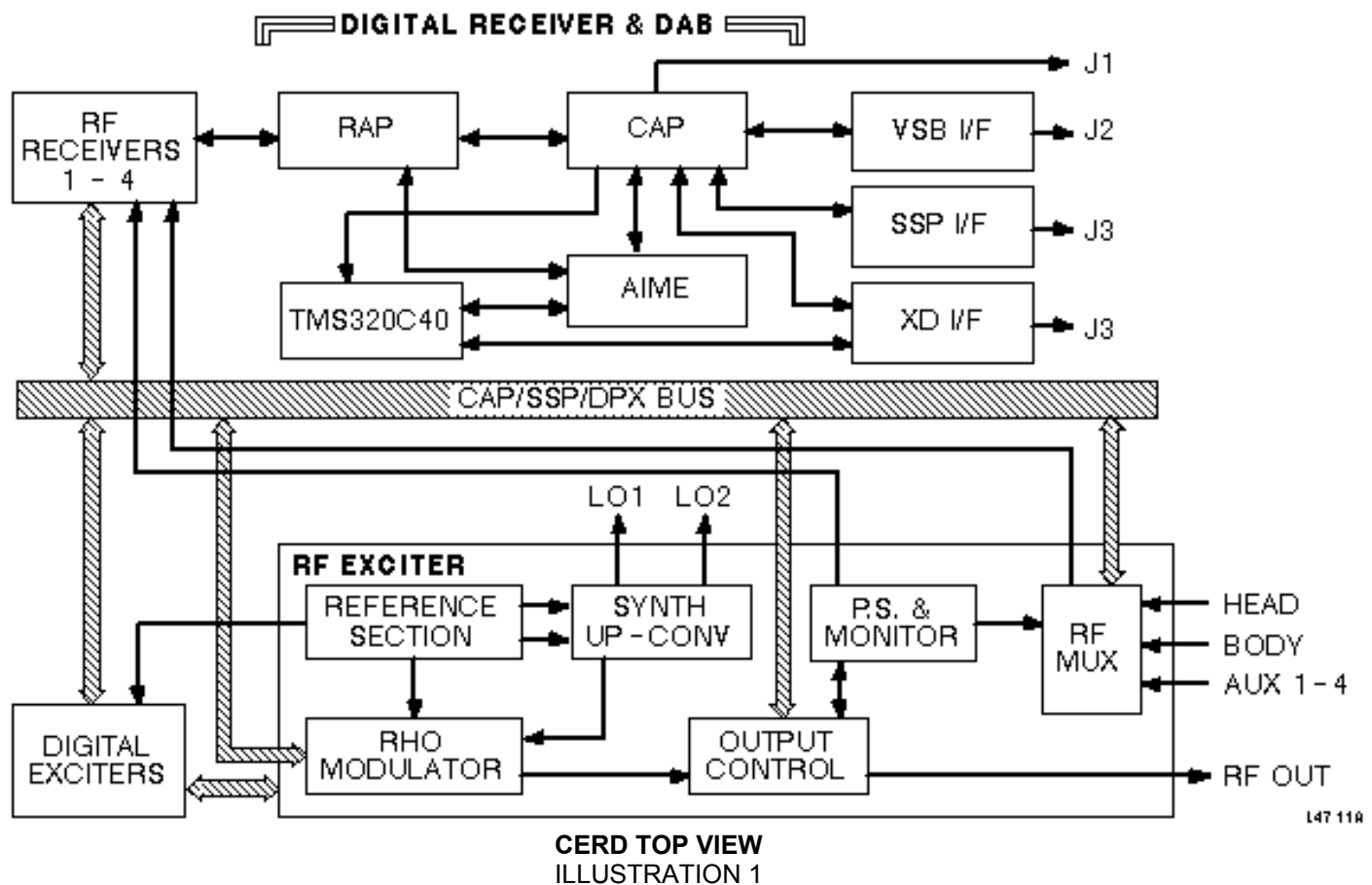
1 - OVERVIEW

The Combined Exciter/Receiver DAB (CERD) is a combination of the Exciter, Receiver(s), and data acquisition board (DAB) in one triple-width VME module that occupies ISE chassis slots 13, 14, and 15. One-channel and four-channel versions are offered in the same physical package.

The CERD module consists of an extended 9U digital circuit board with a similar sized RF module mounted to it.

Upgrade from one to four channels is accomplished by plugging additional single-channel RF RX boards into the RF module.

The CERD provides a major advancement over the previous transceiver by including a high-speed, digital, frequency-domain processing chip set. Thus, the row FFT portion of reconstruction can be performed on the CERD. The CERD also has a mode that provides time-domain data to allow compatibility with current image reconstruction software.



2 - RECEIVERS

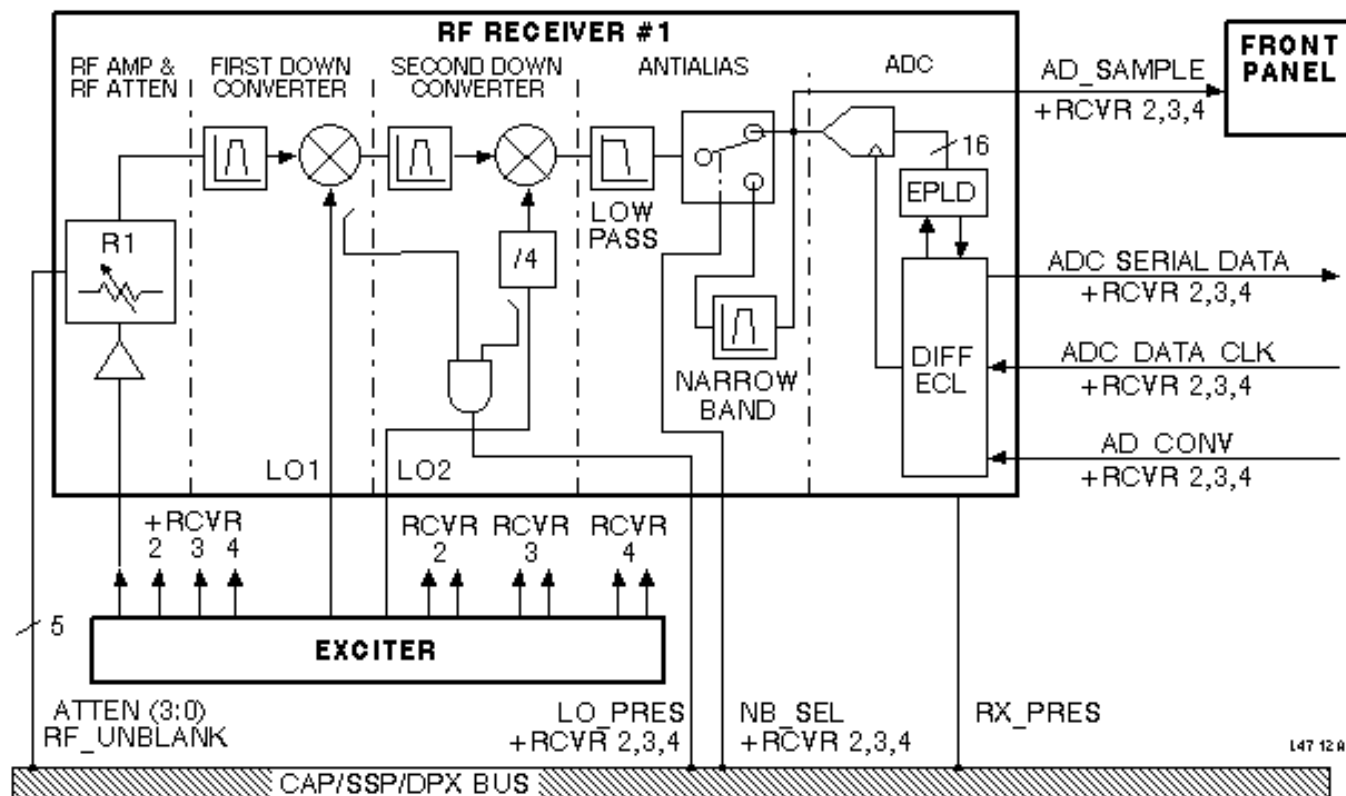


TABLE 1
RECEIVER #1 CONNECTOR

Pin #	Signal Name	Signal Flow	Signal	Description
1	N/C			
2	N/C			
3	RX1_AD_CAL	Digital RF	Note 2	ADC calibrate
4	RX1_PRESENT	RF Digital	Note 2	Rx channel present
5	RX1_3dB	Digital RF	Note 2	R1 3dB
6	RX1_12dB	Digital RF	Note 2	R1 12dB
7	RX1_LO1_PRES	RF Digital	Note 2	LO1 present
8	LGND	Digital RF	GND	Logic GND through 1K pulldown
9	RX1_AD_CONV-N	Digital RF	Note 1	ADC convert strobe –
10	LGND	Digital RF	GND	Logic GND through 1K pulldown
11	RX1_DATA-N	RF Digital	Note 1,3	ADC serial data out –
12	LGND	Digital RF	GND	Logic GND through 1K pulldown
13	RX1_DATA_CK-N	Digital RF	Note 1	ADC serial data out clock –
14	N/C			
15	N/C			
16	RX1_UNBLNK	Digital RF	Note 2	RF module unblank
17	RX1_6dB	Digital RF	Note 2	1 6dB
18	RX1_24dB	Digital RF	Note 2	R1 24dB
19	RX1_NB_SEL	Digital RF	Note 2	Select narrowband filter
20	LGND	Digital RF	GND	Logic GND through 1K pulldown
21	RX1_AD_CONV-P	Digital RF	Note 1	ADC convert strobe +
22	LGND	Digital RF	GND	Logic GND through 1K pulldown
24	LGND	Digital RF	GND	Logic GND through 1K pulldown
25	RX1_DATA_CK-P	Digital RF	Note 1	ADC serial data out clock +

Note 1: PECL (+5V supply differential ECL) with 100 parallel termination

Note 2: TTL with input pulldown resistor

Note 3: Output of differential receiver on digital board guaranteed to be logic 0 if RF receiver disconnected

Receiver #2 Connector: Same as Receiver #1 except replace all instances of "RX1" with "RX2"

Receiver #3 Connector: Same as Receiver #1 except replace all instances of "RX1" with "RX3"

Receiver #4 Connector: Same as Receiver #1 except replace all instances of "RX1" with "RX4"

3 - RF RECEIVER INTERFACE

In general, the firmware interface with the receiver functions may take one of the following forms:

- CAP/SSP access
- CAP-only access
- ASC-only access

4 - EXCITERS

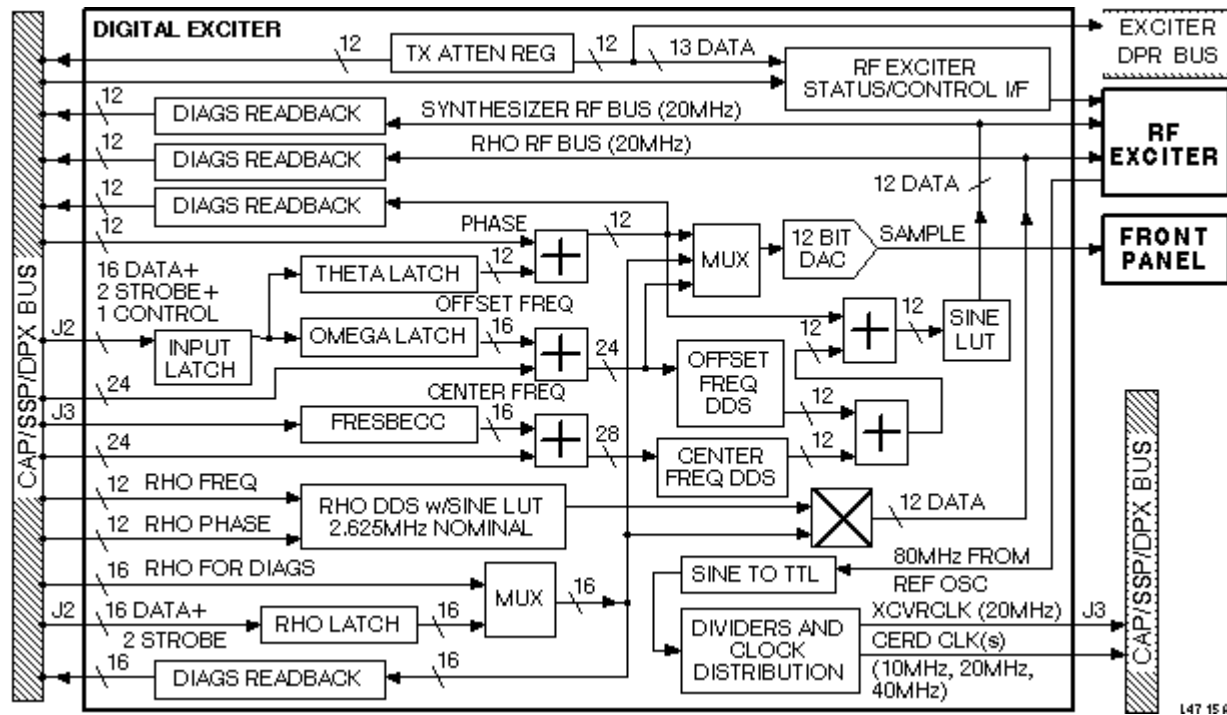


TABLE 2
FRONT PANEL: RF EXCITER BOARD CONNECTORS

Jumper #	Signal Name/Function	Connector Type	Signal type
201	A/D Sample 1	BNC	Analog 5V (1K series resistor)
202	A/D Sample 2	BNC	Analog 5V (1K series resistor)
203	A/D Sample 3	BNC	Analog 5V (1K series resistor)
204	A/D Sample 4	BNC	Analog 5V (1K series resistor)
17	RF Sample Out	BNC	50 ohm out
18	LO1 Out	BNC	50 ohm out
19	LO2 Out (CERD Spectro Exciter)	BNC	50 ohm out
20	Analog power	25-pin sub-D, male	
21	RF Out to PA, Body, Head, AUX1, AUX2, AUX3, AUX4, +15VDC	RF coax D-shell, 8-pos.	

TABLE 3
FRONT PANEL: RF EXCITER BOARD POWER CONNECTOR

Power Supply Voltages	Pin-out* Test Points
+5VDC:	1,2,14,15
-5.2VDC:	5,6,18,19
+15VDC:	9,10,11,12
-15VDC:	22,23
GND:	3,4,7,8,13,16,17,20,21,24,25
* Note: 25-pin sub-D connector	



DIGITAL EXCITER
ILLUSTRATION 5

TABLE 4
FRONT PANEL: DIGITAL BOARD CONNECTORS

Junction #	Signal Name/Function	Connector type	Signal Type
4	CAP Programmable	SMB	TTL out (1K series resistor)
5	RX Data In Window	SMB	TTL out (1K series resistor)
6	RF EX Unblank* (was TR Driver Unblank)	BNC	TTL out (1K series resistor)
7	RF RX Unblank*	BNC	TTL out (1K series resistor)
8	CAP Serial Port	Telephone jack RS 232 (6-pos. female)	
9	RHO/THETA/OMEGA Sample	BNC	Analog 5V (1K series resistor)
10	DABOUT 6 (= old DAB J10)	BNC	TTL out (1K series resistor)
11	DABOUT 7 (= old DAB J11)	BNC	TTL out (1K series resistor)
12	DABOUT 8 (= old DAB J12)	BNC	TTL out (1K series resistor)

The following signals currently available on the DAB are not available on the CERD front panel:

- DAB J5: DABOUT 1 (EFB EN*); available on TYME II Board
- DAB J6: DABOUT 2 (LOCKOUT*); available on TYME II Board
- DAB J7: DABOUT 3 (CINEFILT*); available on TYME II Board
- DAB J8: DABOUT 4 (SPECTRO UNBLANK); available on TYME II Board
- DAB J9: DABOUT 5 (SCOPE TRIG); available on IPG

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 2, 1998	J. Saperstein	Initial Conversion from Toolbook to Word.
1	Oct 13, 1999	M. Keber	Added correct proprietary heading to document.